

Indian Institute of Technology, Gandhinagar

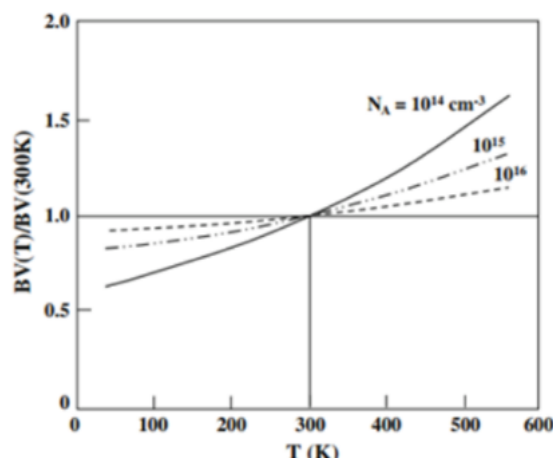
ES226 Semiconductor Devices

Semester 2, Tutorial - 6

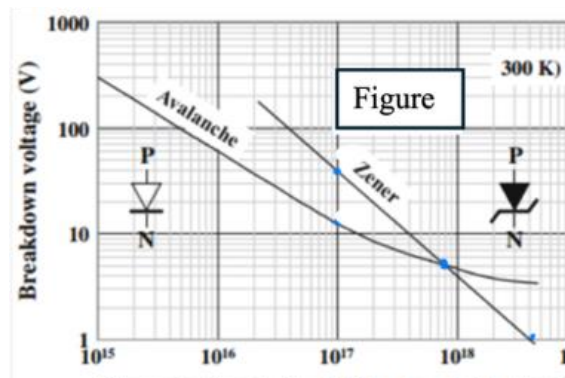
Q1. Describe Breakdown Mechanism in PN junction diode.

- a. Avalanche breakdown. How can we increase Avalanche breakdown voltage?
- b. Zener breakdown. How can we decrease Zener breakdown voltage?
 1. On which parameters does Tunnelling probability depend on?
 2. Does Avalanche breakdown occur in Zener diodes?
- c. Are both these breakdown mechanisms reversible? Explain.
- d. If a diode is given to you randomly, can you go into the lab and find out if it is Zener diode or Avalanche diode?
- e. If the material band gap is increased, how does the Avalanche and Zener Breakdown voltage change?

Q2. The figure given below shows the normalized Breakdown Voltage (BV) for a N^+P junction as a function of temperature (T) with acceptor doping concentration as a parameter. The temperature coefficient of BV changes with doping. Explain the graph with a valid reason.



Q3: The figure given below compares the Zener and Avalanche breakdown voltage (BV) of an N^+P or P^+N junction as a function of the concentration in the lighter doped side. When BV is above 8V, the avalanche multiplication dominates. When BV is less than $\sim 4 \text{ V}$, the tunnelling mechanism dominates. Why? Explain your answer with proper reason.



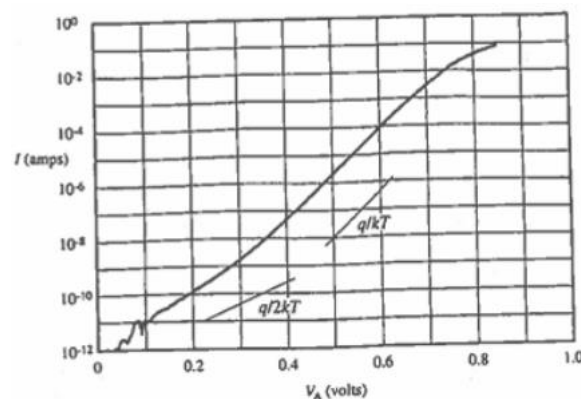
Q4. Consider a silicon n^+p junction diode. The critical electric field for breakdown in silicon is approximately $E_{\text{crit}} = 4 \times 10^5 \text{ V/cm}$. Determine the maximum p-type doping concentration such that the breakdown voltage is (a) 40 V and (b) 20 V.

Q5. Suppose in the I-V characteristics of two tunnel diodes, the I_p of Diode 1 is reached at a V_p which is lower than that of Diode 2. What can we comment on the positions of E_F . Explain.

Q6. How do the forward-bias (FB) and reverse-bias (RB) I-V characteristics change when the semiconductor material is varied? Additionally, how does the choice of semiconductor material affect the breakdown voltage?

Q7. How do the forward-bias (FB) and reverse-bias (RB) I-V characteristics of a silicon diode change with temperature? Additionally, how does temperature affect the breakdown voltage?

Q8. Based on the given plot estimate the series resistance (R_s) in the diode. Assume the q/kT slope cuts 0.1A at 0.78V.



Q9. Describe tunnel (Esaki) diode operation in forward and reverse bias. Why do you get negative resistance and what does it signify?

Q10. How does increasing doping concentration beyond the optimum value affect the negative resistance region of a tunnel diode?

Q11. A diode shows breakdown at 4.5 V at 25 °C. When temperature increases to 75 °C, breakdown voltage reduces to 4.2 V. What can you infer about the breakdown mechanism? Also, what will happen if doping concentration is reduced?

Q12. Plot the small signal resistance R_p as a function of applied voltage for V_a positive. Let the reverse leakage current be $I_0 = 10^{-15} \text{ A}$. Use a logarithmic scale for the resistance axis. Comment on your graph in view of your expectations of the resistance of a diode.

Q13. Note that diffusion coefficient, diffusion length and lifetime depend on doping. Consider a one-sided n^+p junction in silicon.

a) Plot the reverse diffusion current density as a function of doping for $V_a = -5 \text{ V}$

b) Plot the reverse generation current density as a function of doping for $V_a = -5 \text{ V}$.

Q14. A Si junction has $N_D' = 10^{15} \text{ cm}^{-3}$ and $N_A' = 10^{17} \text{ cm}^{-3}$. The junction area is $100 \mu\text{m}^2$.

a) Find V_{bi}

b) Find I_0 .

c) Find the current at $V_a = -5\text{ V}$, 0 V , and $+0.5\text{ V}$. Remember to consider both diffusion and recombination-generation current. Also bear in mind that most of the junction appears on the lightly doped side (thus lifetime should be chosen for that material).

Q15. An abrupt Si p-n junction ($A = 10^{-4}\text{ cm}^2$) has the following properties at 300 K :

p side: $N_a = 10^{17}\text{ cm}^{-3}$, $\tau_n = 0.1\text{ }\mu\text{s}$, $\mu_p = 200\text{ cm}^2/\text{V-s}$, $\mu_n = 700\text{ cm}^2/\text{V-s}$

n side: $N_d = 10^{15}\text{ cm}^{-3}$, $\tau_p = 10\text{ }\mu\text{s}$, $\mu_n = 1300\text{ cm}^2/\text{V-s}$, $\mu_p = 450\text{ cm}^2/\text{V-s}$

The junction is forward biased by 0.5 V . What is the forward current? What is the current at a reverse bias of -0.5 V ?

Q16. Determine the ideal reverse-saturation current density in a silicon pn junction $T = 300\text{ K}$

Consider the following parameters in a silicon pn junction:

$N_a = N_d = 10^{16}\text{ cm}^{-3}$, $n_i = 1.5 \times 10^{10}\text{ cm}^{-3}$, $D_n = 25\text{ cm}^2/\text{s}$, $D_p = 10\text{ cm}^2/\text{s}$, $\tau_p = \tau_n = 5 \times 10^{-7}\text{ s}$, $\epsilon_r = 11.7$

Q17. Consider a symmetrical junction in GaAs in which $N_D' = N_A' = 10^{16}\text{ cm}^{-3}$.

a) Find V_{bi}

b) Calculate the series resistance R_s of the bulk regions if the cross-sectional area of the junction is $50\text{ }\mu\text{m}^2$ and the lengths of the bulk regions are each $2\text{ }\mu\text{m}$. Note that the series resistance R_s is due to the finite resistivity of the semiconductor materials; it is not the same as the differential resistance R_p .